# Distinct-Target Certificates and Quantitative Reductions for Seymour's Second-Neighborhood Problem

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**Abstract**

This report examines the pinned manuscript *A proof of Seymour's second-neighborhood conjecture* and its applicability to the private Ram rainbow-reachability report. We distinguish exact-distance vertex sets from walk counts, a local certificate from its universal existence, and independent clones from transitive-tournament substitution. The report proves an exact two-label specialization, a quantitative deletion-stage identity, a sharper nonuniform substitution estimate, exact clone identities, and robust distinct-target certificates. It reconstructs the manuscript's pruning and extremal dependencies without treating a formalization catalogue entry or finite tests as a global proof certificate. The direct private report is read in full; additional corpus passages are read at their actual source locations. All assigned mathematical source files are read and checked against their stored hashes. A reproducible exact checker and a precise evidence ledger accompany the report. Applicability level L5 denotes direct relevance, not validation of a conjecture's resolution.

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# 1 Scope, evidence classes, and the mathematical target

The public snapshot specified for this study is pinned to

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The source manuscript is dated September 23, 2026. Its main assertion is that every nonempty finite oriented graph has a vertex $v$ with $|N_1^+(v)| \le |N_2^+(v)|$. The word *oriented* means loopless, without parallel arcs and without opposite arcs; it does not mean a tournament. A sink satisfies the inequality with both sides zero.

Four logically different types of evidence will be kept separate:

1. **Source claim.** The manuscript asserts the unrestricted theorem and gives a reduction, a pruning argument, and an extremal contradiction.
2. **Local proof.** A statement proved here from explicitly stated finite hypotheses, such as an identity for a particular blowup or a sufficient certificate for a specified vertex, is independent of acceptance of that claim.
3. **Global conditional.** Consequences of the unrestricted theorem, or of the manuscript's central pruning premise, are presented with that dependency visible.
4. **Finite test.** Exact computations check stated bounded families. They neither establish unbounded existence nor replay a formal theorem.

The catalogue's L5 assignment is correct as an applicability judgment: the private two-identical-label problem specializes exactly to this manuscript. It would be incorrect to infer from L5 that the private local surplus lemma proves the universal existence of a successful vertex. Likewise, a statement in a formalization catalogue is metadata until its actual theorem, assumptions, dependencies and checking environment have been independently examined.

# 2 Exact sets and the private two-label specialization

## 2.1 Image sets versus distances

For $S \subseteq V(D)$ define the ordinary image

$$F(S) = \bigcup_{x \in S} N_1^+(x), \qquad F^2(S) = F(F(S)).$$

The image operator preserves unions but does not remove $S$. For a singleton in an oriented graph there is no length-two return, and hence

$$N_2^+(v) = F^2(\{v\}) \setminus F(\{v\}). \tag{1}$$

For a general set $S$, the expression $F^2(S) \setminus F(S)$ can intersect $S$. For example, on a directed triangle with $S$ a two-vertex set, the missing vertex of $F(S)$ belongs to $S$ and is in $F^2(S)$. There is no permission to replace this expression by the vertices at set-distance exactly two, which would also exclude $S$.

If $A$ is the 0–1 adjacency matrix, then $(A^2)_{vw}$ counts intermediate vertices, not distinct endpoints. The correct count is

$$|N_2^+(v)| = \sum_{w \neq v} (1 - A_{vw}) \mathbf{1}\{(A^2)_{vw} > 0\}. \tag{2}$$

Take arcs $v \to a, v \to b, a \to c, b \to c$ and no others. The first neighborhood has size two and the exact second neighborhood size one, although there are two length-two walks. Even thresholded $A^2$ must have the first neighborhood removed; in a transitive triangle its only two-step endpoint is already a first neighbor.

### 2.2 Specialization with no conjectural input

The private report [2] uses a label palette $S_e \subseteq [k]$ on each arc, distinct labeled first neighbors $A(v)$, surplus $\sigma(v) = \sum_{vx}(|S_{vx}| - 1)$, and new rainbow two-step endpoints $B(v)$. The load identity is $\sum_i d_i^+(v) = |A(v)| + \sigma(v)$. Rainbow paths choose different labels at different steps, and reachability uses positive-length paths, not the identity term of a closure matrix.

**Proposition 2.1** (Exact Seymour slice)**.** *Let $D$ be a nonempty finite oriented graph, give every arc the palette $\{1, 2\}$, and let $\mathcal{R}(v)$ be its rainbow-reachable vertices. Then*

$$A(v) = N_1^+(v), \quad \sigma(v) = |N_1^+(v)|, \quad B(v) = N_2^+(v), \quad \mathcal{R}(v) = N_1^+(v) \,\dot\cup\, N_2^+(v).$$

*There is no rainbow directed cycle. The rainbow reachability inequality at $v$ is equivalent to the second-neighborhood inequality at that same vertex.*

*Proof.* A rainbow path has at most two arcs because only two labels are available. Every one-arc path and every two-arc directed path can be labeled, using 1 and 2 in either order in the latter case. An oriented graph has no loop or two-cycle, so a two-step walk from $v$ is a path and cannot return to $v$. Its endpoint is either already a first neighbor or belongs to $N_2^+(v)$. These facts give all four identities. Every directed cycle has length at least three, too many arcs for the palette. Finally, $|\mathcal{R}(v)| \geq \sum_i d_i^+(v)$ becomes $d_1(v) + d_2(v) \geq 2d_1(v)$. □

This is a specialization of definitions, not a new resolution of the general multi-label conjecture. Sullivan explicitly records the identical two-class specialization in his public report [6]. The private report's surplus theorem supplies a correct *sufficient* local test; in this slice its test is exactly the desired inequality and does not prove that some vertex passes it.

Other correct private structural results become weaker in this slice. Its multi-label core is exactly $D$, so the acyclic-core theorem says an acyclic digraph has a sink. The common two-palette condition on a core triangle is automatic, because every arc already has palette $\{1, 2\}$. Higher cores $M_q$, $q \geq 3$, are empty. These facts preserve the private results but do not convert them into the manuscript's subset deficit or its pair-family pruning theorem.

### 2.3 A matching test and its strict limitation

In the two-label slice the private debt-token graph has one token for each $x \in N_1^+(v)$, joined to $y \in N_2^+(v)$ when $x \to y$. Either choice of base label gives the same incidence because the second arc has both labels. A matching covering all tokens provides an injective endpoint certificate. Hall's condition on this bipartite graph is sufficient, but stronger than $d_2(v) \geq d_1(v)$.

To prove the distinction, take only arcs $v \to x, v \to z, x \to a, x \to b$. Then $d_1(v) = d_2(v) = 2$, while the token at $z$ has no neighbor. The maximum matching has size one. A failed token allocation is therefore not even a certificate of failure at a specified vertex. Enlarging an incidence relation is valid only when each new edge still records an actual path with legal labels.

**Proposition 2.2** (Capacitated local certificate)**.** *Fix a finite oriented graph, $v \in V$, and weights $\eta : V \to [0, \infty)$. Suppose numbers $f_{xy} \geq 0$, indexed by $x \in N_1^+(v)$ and $y \in N_2^+(v)$ with $x \to y$, satisfy*

$$\sum_{y:x\to y} f_{xy} \geq \eta(x) \quad (x \in N_1^+(v)), \qquad \sum_{x:x\to y} f_{xy} \leq \eta(y) \quad (y \in N_2^+(v)).$$

*Then $\eta(N_2^+(v)) \geq \eta(N_1^+(v))$. Moreover, such numbers exist if and only if*

$$\eta(X) \leq \eta(\Gamma(X)) \quad \textit{for every } X \subseteq N_1^+(v),$$

*where $\Gamma(X) = N_2^+(v) \cap F(X)$.*

*Proof.* Summing the row and column inequalities proves the first assertion. Necessity of the family inequalities follows by summing just the rows in $X$. For sufficiency form a network from a source to each $x$ with capacity $\eta(x)$, from each allowed $x$ to $y$ with capacity $1 + \eta(N_1^+(v))$, and from each $y$ to the sink with capacity $\eta(y)$. Every cut using a middle edge is larger than the total required flow. Every other cut, whose source-side left vertices form $X$, has capacity at least $\eta(N_1^+(v) \setminus X) + \eta(\Gamma(X))$, at least the required total. The finite max-flow/min-cut theorem gives a saturating flow. It applies to real capacities; no integrality or rounding is needed for this existence assertion. With integer capacities an integral flow is available. □

# 3 Counterexample reduction and quantitative subset deficits

## 3.1 What minimality permits

Suppose, conditionally, that a counterexample exists. Choose $D$ of minimum order and then minimum number of arcs among all counterexamples. It has no sink. A sink strongly connected component has all its first and second out-neighborhoods internally, so its induced graph would still be a counterexample. Minimum order therefore forces strong connectivity and positive indegrees. This is a valid induced-component reduction, not arbitrary iterative deletion.

The private sink-core reduction is compatible with this argument. For a general palette, only multi-label arcs must stay within a sink core component; external one-label arcs add one unit of load and one distinct external endpoint each. In the two-label slice every arc is multi-label, so there are no external arcs at all. The private correction is important: on a single arc $u \to v$ with two labels, deleting $v$ makes $u$ a sink, but originally $u$ reaches only one endpoint against load two.

## 3.2 A stage identity independent of global minimality

Fix an arbitrary oriented $D$, a nonempty proper $S$, and put

$$I = F(S) \cap S, \quad E = F(S) \setminus S, \quad g(T) = |F(T) \setminus (I \cup T)| \quad (T \subseteq E).$$

For $T \subsetneq E$, delete all arcs from $S$ to $E \setminus T$ to form $D_T$. Suppose a particular vertex $v \in S$ loses a nonempty set $Q = N_{1,D}^+(v) \cap (E \setminus T)$. Define

$$C = F(Q) \setminus \big(I \cup T \cup Q \cup F(T)\big), \qquad h(v) = d_{1,D}(v) - d_{2,D}(v).$$

**Lemma 3.1** (Quantitative deletion stage)**.** *With these definitions,*

$$g(T \cup Q) - g(T) = |C| - |Q \cap F(T)|, \tag{3}$$

$$d_{2,D_T}(v) \leq d_{2,D}(v) - |C| + |Q \cap F(T)|. \tag{4}$$

*If* $d_{1,D_T}(v) \le d_{2,D_T}(v)$*, then*

$$g(T \cup Q) - g(T) \le |Q| - h(v). \tag{5}$$

*These assertions do not assume that all vertices of* $D$ *are bad.*

*Proof.* Every surviving first neighbor of $v$ belongs to $I \cup T$. A subsequent arc from $I \subseteq S$ ends in $I \cup T$, and one from $T$ ends in $F(T)$. Thus $F^2_{D_T}(\{v\}) \subseteq I \cup T \cup F(T)$. Deletion creates no walks, so an endpoint newly counted at exact distance two must have been a deleted first neighbor. It lies in $Q \cap F(T)$. Conversely, each vertex in $C$ was a second neighbor through $Q$, was not an original first neighbor, and has no surviving two-step path. It is not $v$: an arc from $Q$ to $v$ would be opposite an original arc from $v$ to $Q$. This proves (4).

In the set defining $g(T)$, adjoining $Q$ removes exactly $Q \cap F(T)$. The new image points that are neither already counted nor excluded are exactly $C$. This proves (3). Finally $d_{1,D_T}(v) = d_{1,D}(v) - |Q|$; insert this into (4) and use the assumed success of $v$ in $D_T$. □

**Corollary 3.2** (Accumulated deficit)**.** *For a minimum-order, minimum-arc counterexample and any nonempty proper* $S$*, the procedure has stages* $Q_1, \dots, Q_t$ *partitioning* $E$*, with witnesses* $v_1, \dots, v_t \in S$*, and*

$$\delta(S) := |F(S) \setminus S| - |F^2(S) \setminus F(S)| \ge \sum_{j=1}^{t} h(v_j) \ge t \ge 1. \tag{6}$$

*Proof.* Strong connectivity ensures $E \neq \varnothing$. Every $D_T$ deletes an arc when $T \neq E$ and has the same order, so arc minimality supplies a successful vertex. A vertex losing no outgoing arc cannot improve, because its exact second neighborhood can only shrink. Thus the successful vertex lies in $S$ and has $Q \neq \varnothing$. Start from $T = \varnothing$, apply the lemma, and append $Q$ until $T = E$. Each stage uses deletion from the original $D$; the witnesses need not be distinct. The sum of (5) is $g(E) \le |E| - \sum_j h(v_j)$. Since $I \subseteq S$ implies $F(I) \subseteq F(S)$,

$$F^2(S) \setminus F(S) = F(E) \setminus (I \cup E),$$

whose size is $g(E)$. All $h(v_j)$ are positive integers. □

This gives the source's strict subset deficit and also preserves the otherwise discarded stage count and vertex margins. It does not provide a nonvacuous example satisfying the global counterexample hypothesis. The checker exercises the stage identity on ordinary graphs where a bad vertex actually becomes good after the indicated deletion, rather than claiming to test nonexistent known minimal counterexamples.

# 4 Sharper transitive substitution estimates

## 4.1 Exact internal curvature

Let $T_m$ have vertices $1, \dots, m$ and arcs $a \to b$ for $a < b$. For a nonempty $T \subseteq [m]$ with least element $a$, its internal image is $f(T) = \{a+1, \dots, m\}$. If $a < m$, then

$$|T| - 2|f(T)| + |f^2(T)| = |T| - (m - a) - 1 \le 0. \tag{7}$$

If $a = m$, then $T = \{m\}$ and this expression equals one. Thus the internal cost of one in the source is needed only for the singleton last vertex. Equality in (7) holds exactly for a full suffix $T = \{a, \dots, m\}$ with $a < m$.

**Theorem 4.1** (Nonuniform quantitative substitution)**.** *Let $D$ be any finite oriented graph and assign each vertex a positive integer $m_v$. Replace $v$ by $T_{m_v}$, and replace each base arc by all arcs between its two blocks. Call the resulting oriented graph $G$. For a nonempty $U \subseteq V(G)$ let $T_v$ be its selected labels, $S = \{v : T_v \neq \varnothing\}$, and*

$$p(U) = |\{v \in S \setminus F(S) : T_v = \{m_v\}\}|.$$

*For $m(A) = \sum_{v\in A} m_v$ one has*

$$\begin{aligned} |U| - 2|F_G(U)| + |F_G^2(U)| \leq m(F^2(S) \setminus F(S)) \\ - m(F(S) \setminus S) + p(U). \end{aligned} \tag{8}$$

*No subset-deficit, strong-connectivity or minimum-degree assumption is needed for this estimate.*

*Proof.* Write $H = F(S) \cup F^2(S)$. Blocks in $F(S)$ are filled by the first image; the other nonempty first-image blocks contain exactly $f_v(T_v)$. Hence

$$|F_G(U)| = m(F(S)) + \sum_{v\in S\setminus F(S)} |f_v(T_v)|.$$

A two-step path with two external arcs ends in an $F^2(S)$ block. With one external arc it ends in an $F(S)$ block. With no external arc it contributes $f_v^2(T_v)$. Therefore

$$|F_G^2(U)| \leq m(H) + \sum_{v\in S\setminus H} |f_v^2(T_v)|.$$

In the curvature expression, enlarge the last sum to $S \setminus F(S)$; all added terms are nonnegative. Bound $|T_v| \leq m_v$ for $v \in S \cap F(S)$ and apply (7) on the other blocks. The result is at most

$$m(H) - 2m(F(S)) + m(S \cap F(S)) + p(U),$$

which is the right side of (8) after disjoint-set decomposition. Repeated two-step paths into one block have not been counted as separate endpoints anywhere in this proof. □

**Corollary 4.2** (Reduced uniform threshold)**.** *Suppose $D$ has $n$ vertices, positive indegrees, and $\delta(S) \geq 1$ for every nonempty proper $S$. A uniform block size $m$ gives strict negative curvature for every nonempty proper $U$ provided*

$$m\delta(S) > |S \setminus F(S)| \quad (\varnothing \neq S \subsetneq V(D)). \tag{9}$$

*In particular $m = n$ suffices, improving the manuscript's sufficient condition $m > n$. More precisely the sufficient integer threshold is*

$$1 + \left\lfloor \max_{\varnothing\neq S\subsetneq V(D)} \frac{|S \setminus F(S)|}{\delta(S)} \right\rfloor.$$

*Proof.* For proper $S$, (8) is at most $-m\delta(S) + p(U) \leq -m\delta(S) + |S \setminus F(S)|$. For $S = V(D)$, positive indegrees give $F(S) = V(D)$ and fill all blocks in one step. Positive indegrees also hold in $G$, so $F_G(U) = F_G^2(U) = V(G)$, and the curvature is exactly $|U| - |V(G)| < 0$. Finally, for proper $S$, $|S \setminus F(S)| \leq n - 1$. The hypotheses exclude $n = 1$, since a loopless one-vertex graph has indegree zero. □

Nonuniform block design is possible as well: it suffices to choose positive integers with

$$m(F(S) \setminus S) - m(F^2(S) \setminus F(S)) > |S \setminus F(S)|$$

for all proper nonempty supports. This is a finite integer feasibility condition, not an assertion that unequal sizes always improve the optimum. The margin in (6) can lower the uniform threshold when stage information is available. No optimal block-size claim is made.

# 5 Pruning: source reconstruction and proof obligations

## 5.1 The survivor-dependent bound

For a finite relation on $X$, consider separate left and right families $\mathcal{R}, \mathcal{C} \subseteq X^2$. A left pair $(p, j)$ conflicts with a right pair $(i, s)$ when $p \to i$ and $s \to j$. The two corners are

$$Z = \{(i, j) : \text{a conflict witnesses this corner}\}, \qquad H = \{(p, s) : \text{a conflict witnesses this corner}\}.$$

The manuscript's pruning lemma asserts the existence of conflict-free subfamilies retaining every initially conflict-free member, with deletion cost at most $|H| + q$, where $q$ counts corners in $Z$ left uncovered *by the retained families*. Left members cover $(i, j)$ through their first-coordinate arrow; right members cover it through their second-coordinate arrow. Copies of an ordered pair on opposite sides are different objects.

This is stronger than a bound using the original number of conflicts or the number of points initially uncovered. The post-deletion term is exactly what the subsequent extremal objective needs. The lemma is stated for all finite binary relations, including loops and opposite arrows, not only oriented graphs. The finite checker accordingly includes such relations.

## 5.2 Rank facts that can be proved locally

**Lemma 5.1** (Maximum-support-rank kernel test)**.** *Let $D : \mathbb{R}^J \to \mathbb{R}^I$ attain the largest rank among all real matrices supported on $T \subseteq I \times J$. For $x \in \ker D$, $y \in \ker D^T$ and $(s, j) \in T$, one has $x_j y_s = 0$.*

*Proof.* If $x_j y_s \neq 0$, for any $\tau \neq 0$ set $D_\tau = D + \tau e_s e_j^T$. Then $D_\tau x = \tau x_j e_s$, so $e_s \in \operatorname{im} D_\tau$. Also $Dw = D_\tau w - \tau w_j e_s$, giving $\operatorname{im} D \subseteq \operatorname{im} D_\tau$. But $y^T e_s \neq 0$ whereas $y^T D = 0$, so $e_s \notin \operatorname{im} D$. Thus the perturbed, still-supported matrix has larger rank, a contradiction. □

The perturbation need not preserve any other matrix's rank. It is a contradiction to maximality for this one support problem. This detail prevents an incorrect demand that all simultaneous genericity conditions survive the hypothetical perturbation.

The four maps in the source have entries

$$L_{(i,j),(p,j)} = a_{pi}, \qquad N_{(i,s),(i,j)} = b_{sj},$$
$$B_{(p,s),(p,j)} = b_{sj}, \qquad A_{(i,s),(p,s)} = a_{pi},$$

where the spaces are respectively $L : \mathbb{R}^{\mathcal{R}} \to \mathbb{R}^Z$, $N : \mathbb{R}^Z \to \mathbb{R}^{\mathcal{C}}$, $B : \mathbb{R}^{\mathcal{R}} \to \mathbb{R}^H$ and $A : \mathbb{R}^H \to \mathbb{R}^{\mathcal{C}}$. Entries outside the relation support vanish.

**Proposition 5.2** (Exact corner factorization)**.** *For every coefficient assignment, $NL = AB$.*

*Proof.* For a fixed column $(p, j)$ and row $(i, s)$, the only possible intermediate index of $NL$ is $(i, j)$; for $AB$ it is $(p, s)$. If the endpoints conflict, both indices are present and both products are $a_{pi} b_{sj}$. Otherwise an arrow factor is zero in each product. The equality is entrywise; it does not compare generic matrix ranks by analogy. □

## 5.3 The full source dependency chain

The entire pruning proof was read. Its substantive steps, including the quantifiers that a verification must preserve, are as follows.

1. Augment the conflict graph by separate copies $Z_L, Z_R$. A conflict creates a path from its $Z_L$ corner through its two original members to its $Z_R$ corner. Different conflicts may share this path's vertices. Initially conflict-free members are isolated even after augmentation.

2. Choose a maximum matching with the fewest original-to-original edges. Writing its counts as $k, \ell, t$, the two side matchings are individually maximum: an augmenting path would either increase the total size or replace one original edge and reduce $k$. This lexicographic choice cannot simply be omitted.

3. Choose coefficients making the two matching minors nonzero and each restricted matrix $D^{p,i} = (b_{sj})_{s\in S_i, j\in J_p}$ attain its maximum support rank. Within a coordinate block, allowed entries are independent variables. Variables shared *between* blocks do not invalidate the argument: the product of finitely many nonzero determinant polynomials is nonzero over $\mathbb{R}$. Empty minors are one.

4. Basis columns yield lifts $u^r \in \ker L$ and $v^c \in \ker N^T$ with unit coordinates at their respective unmatched-by-side original endpoints. For every $u \in \ker B$ and $v \in \ker N^T$, restrictions lie in the two kernels of $D^{p,i}$. A missing row or column is either all zero or would itself witness the missing corner, so no needed equation is lost. Lemma 5.1 then gives $u_{pj}v_{is} = 0$ at every conflict.

5. Substituting each lifted $v^c$ forces all vectors of $\ker B$ to vanish at the corresponding original left endpoint. Hence the $Bu^r$ are independent and lie in $\ker A$ by Proposition 5.2. This yields

$$|H| \geq k + \operatorname{rank} A \geq k + \operatorname{rank}(NL) \geq k + \ell + t - |Z|.$$

6. Alternating paths produce a vertex cover of size at most $|H| + |Z|$. Every corner of $Z$ still covered after deletion requires a corresponding copy in the cover. Subtracting these $|Z| - q$ copies gives deletion cost at most $|H| + q$, and isolated original members survive.

No explicit algebraic counterexample to these read steps was located in this examination. This reconstruction is not a Lean replay or independent certification of the manuscript's unrestricted theorem. In the global discussion below the pruning assertion is kept as a named source premise. Our exact tests verify the actual augmented matching bound, the extracted cover, survivor compatibility, preservation of isolated members and the post-deletion charge on the specified finite families; they do not infer the rank theorem from random numerical ranks.

## 6 The extremal step, with its global dependency exposed

Assume a nonempty oriented $G$ has positive indegrees and strict curvature $|U| + |F^2(U)| < 2|F(U)|$ for every nonempty proper $U$. Let $F_1, F_2$ advance the first and second coordinates of subsets of $X^2$, and let $\Delta = \{(x, x) : x \in X\}$. Choose compatible $P, Q$ containing $\Delta$ in both, where compatibility means $F_1P \cap F_2Q = \varnothing$. The choice $P = Q = \Delta$ is possible precisely because there are no opposite arcs or loops. Maximize

$$M + d, \qquad M = |P| + |Q|, \quad d = |X|^2 - |F_1P| - |F_2Q|.$$

Every column $P_j$ and row $Q_i$ is nonempty because of the diagonal. It is proper: $F(X) = X$ by positive indegrees, and a full column or row would conflict with an appropriate diagonal image

using an in-neighbor. Thus the strict curvature inequality is applicable fiber by fiber. Summing it and taking complements gives prospective families

$$\mathcal{R} = X^2 \setminus F_2^2 Q, \qquad \mathcal{C} = X^2 \setminus F_1^2 P, \qquad |\mathcal{R}| + |\mathcal{C}| > M + 2d.$$

The diagonal remains present and conflict-free: a purported diagonal conflict traces a two-step path from another diagonal point into the forbidden second image.

For a conflict $((p, j), (i, s))$ of the prospective families, its corner $(p, s)$ lies outside $F_1 P \cup F_2 Q$. If it were in $F_1 P$, append $p \to i$ to reach $(i, s) \in F_1^2 P$; if it were in $F_2 Q$, append $s \to j$ to reach $(p, j) \in F_2^2 Q$. Both contradict the complements. Consequently $|H| \leq d$. Applying the source pruning premise leaves compatible $P', Q'$ still containing the diagonal and deletes at most $d + d'$ members, where $d'$ is their total number of uncovered points. Then

$$|P'| + |Q'| + d' > M + 2d - (d + d') + d' = M + d,$$

a contradiction.

This verifies the algebra of the extremal implication conditional on the pruning premise; in particular the sign of $d'$ and the preservation of both diagonal copies are essential. Combining it with the conditional minimal-counterexample reduction is the manuscript's claimed proof, not a consequence of the private surplus lemma alone. Replacing the pruning bound by a conventional conflict count would not justify the displayed increase.

# 7 Independent clones, weights, and stable local certificates

## 7.1 A different blowup with an exact neighborhood formula

**Theorem 7.1** (Independent-clone identity)**.** *Let $D$ be finite and oriented, and let $m_v \geq 1$ be integers. Replace $v$ by an independent set $C_v$ of $m_v$ vertices and place all arcs from $C_v$ to $C_w$ exactly when $v \to w$. For any clone $x \in C_v$,*

$$N_1^+(x) = \coprod_{w \in N_{1,D}^+(v)} C_w, \qquad N_2^+(x) = \coprod_{w \in N_{2,D}^+(v)} C_w.$$

*Proof.* Every arc crosses to a base outneighbor, proving the first formula. A two-step path projects to $v \to u \to w$. It cannot return to $v$ by orientation. Conversely every base two-step path lifts to every chosen clone of its endpoint since the intermediate block is nonempty. Removing first-neighbor blocks gives exactly the second formula. □

Independent clones represent integer endpoint weights exactly. Transitive blocks do not: a clone inside a transitive block has internal first neighbors, and mixed internal/external paths affect its images. The two constructions have different purposes and cannot be interchanged. Zero clone counts require special care because deleting an intermediate vertex can destroy a path to a positive-weight target. We avoid this by using positive approximating weights.

**Proposition 7.2** (Conditional universal vertex-weight transfer)**.** *If the unweighted second-neighborhood assertion holds for every nonempty finite oriented graph, then for every such $D$ and every $\eta : V(D) \to [0, \infty)$ some $v$ satisfies $\eta(N_2^+(v)) \geq \eta(N_1^+(v))$.*

*Proof.* For positive rational weights multiply by a common denominator and use Theorem 7.1. A successful clone projects to a suitable base vertex. For arbitrary nonnegative $\eta$, set $\eta_q(v) =$

$(\lfloor q\eta(v)\rfloor + 1)/q$. These are strictly positive rationals and converge coordinatewise to $\eta$. Choose a successful $v_q$ for each. Since $V(D)$ is finite, one vertex occurs along an infinite subsequence; passing to its limit proves the claim. Neighborhoods throughout are those of the fixed original graph, even at vertices of zero weight. □

### 7.2 A robustness bound with an exact domain

Define the signed margin $\mu_\eta(v) = \eta(N_2^+(v)) - \eta(N_1^+(v))$.

**Proposition 7.3** (Weight-error budget)**.** *For real weight vectors $\eta, \widetilde{\eta}$ on a fixed finite oriented graph,*

$$\begin{aligned}|\mu_\eta(v) - \mu_{\widetilde{\eta}}(v)| &\le \sum_{x \in N_1^+(v) \cup N_2^+(v)} |\eta(x) - \widetilde{\eta}(x)| \\ &\le (d_1(v) + d_2(v))\|\eta - \widetilde{\eta}\|_\infty.\end{aligned}$$

*Thus a positive approximate margin larger than this error is a rigorous local certificate. Since $d_1(v) + d_2(v) \le n - 1$, $(n-1)\varepsilon$ is a graph-independent error budget when the coordinate errors are at most $\varepsilon$.*

*Proof.* Subtract the two signed sums and apply the triangle inequality. The two neighborhoods are disjoint and neither contains $v$. □

This proves stability of a certificate already present; it supplies neither an approximate algorithm for every input nor the existence of positive slack. At equality, an error interval crossing zero is inconclusive.

### 7.3 The two quantifier orders and reversal

Let $M_{vs}$ be 1 on exact second neighbors, $-1$ on first neighbors, and zero elsewhere. The source distinguishes

$$\forall \eta \ge 0\ \exists v:\ (M\eta)_v \ge 0 \quad \text{from} \quad \exists \eta \ge 0,\ \mathbf{1}^T\eta = 1:\ M\eta \ge 0.$$

They are not interchangeable by moving quantifiers. In a finite oriented graph reversing every arc gives $M_{\overleftarrow{D}} = M_D^T$: a reversed two-step path corresponds exactly to an original reversed pair, and the one-step exclusion transposes as well.

For completeness, the finite alternative behind the second form is this: either the compact convex set $M\{\eta \ge 0 : \mathbf{1}^T\eta = 1\}$ meets the nonnegative orthant, or strong separation gives $\xi \ge 0$ with $M^T\xi < 0$ in every coordinate. The sign of $\xi$ follows because the orthant is an unbounded cone, and strict negativity follows by testing each simplex vertex. Hence the first weighted assertion on the reversed graph excludes the alternative and yields the second form. Existence for all graphs here remains conditional on Proposition 7.2's unrestricted premise. The allowed mass vector need not be strictly positive.

The same alternative has an exact quantitative formulation. Write $\Sigma = \{x \in \mathbb{R}^n : x \ge 0,\ \mathbf{1}^T x = 1\}$. Then

$$\gamma(D) := \max_{\eta \in \Sigma} \min_v (M\eta)_v = \min_{\xi \in \Sigma} \max_s (M^T\xi)_s. \tag{10}$$

Indeed, the left side is the linear program maximizing $t$ with $M\eta \ge t\mathbf{1}$, $\eta \in \Sigma$. Its dual minimizes $z$ with $M^T\xi \le z\mathbf{1}$, $\xi \in \Sigma$. Both are feasible and bounded, so finite linear-program duality gives equality; compactness of the simplices gives attainment. This is classical minimax, not a new graph existence theorem. A simultaneous nonnegative weighting exists precisely when $\gamma(D) \ge 0$. If

$\gamma(D) < 0$, an attaining $\xi$ gives the strict reversed-graph certificate $M^T\xi \le \gamma(D)\mathbf{1} < 0$. Conversely, any strictly negative $M^T\xi$ forces $\gamma(D) < 0$.

Because $M$ is integral, the feasible simplex intersection is a rational polytope. When nonempty it has a rational extreme point, permitting an exact rational certificate. One may similarly restrict $t, z$ to $[-1, 1]$ for optimal solutions and obtain rational optimal primal and dual vertices. A small negative numerical residual is not such a certificate. Uniformly over vertices,

$$\|M\eta - M\widetilde{\eta}\|_\infty \le \|\eta - \widetilde{\eta}\|_1.$$

Consequently an all-row margin at least $\delta > 0$ survives an $\ell^1$ perturbation of size at most $\delta$, provided the perturbed vector remains an admissible nonnegative mass vector. At zero margin no generic positive perturbation budget follows.

The source's arc-weighted form uses

$$\beta_v(s) = \max\big(\{0\} \cup \{w(us) - w(vs) : v \to u \to s\}\big),$$

with $w(vs) = 0$ for absent arcs, and asserts $\sum_s \beta_v(s) \ge \sum_{u \in N_1^+(v)} w(vu)$ at some vertex. This is the established Seacrest equivalence [8], invoked globally, not proved by the independent-clone formula for arbitrary arc weights. Present arcs of weight zero remain paths. When $w(us) = \eta(s)$ on each present arc, first neighbors contribute zero and exact second neighbors contribute $\eta(s)$; this recovers vertex weights without counting multiple paths. The public proof's auxiliary construction need not preserve tournament structure, so equivalence over all oriented graphs is not automatically equivalence inside a restricted class.

# 8 What the Ram representation does and does not transfer

## 8.1 A four-layer exact state model

For the two-label slice the private state space is $V \times \mathcal{P}(\{1, 2\})$. Transitions from $(x, I)$ to $(y, I \cup \{j\})$ require $x \to y$ and $j \notin I$. Every transition strictly increases the label-set size, so its adjacency operator $T$ satisfies $T^3 = 0$. In the Boolean semiring the reachable-state closure is $I \vee T \vee T^2$, where addition is OR. Over nonnegative integers the same finite sum counts labeled walks; only entrywise positivity recovers support. A two-step base path has two label orders, so integer row sums overcount even before different paths coalesce.

Over a field, $(I - T)^{-1} = I + T + T^2$ is a valid terminating inverse. This agrees with the actual terminating-inverse theorem in core-pat-5 [4] and the strict-degree-lowering criterion in the Volume 1 foundation passage [5]. It is the *used-label grading* that proves nilpotence, not an arbitrary ordering of the vertices of $D$. On a directed triangle the original adjacency satisfies $A^3 = I$ and cannot become nilpotent by a permutation. This explicit example corrects any unrestricted triangularization reading of the graph-transform tier in TR190 [3] while preserving its correctly conditional finite inverse.

## 8.2 Support preservation is an additional obligation

An invertible coordinate change preserves linear algebra, but not generally the entrywise positive-support relation. For instance

$$P = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad S^{-1}PS = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}.$$

The transformed entries have signs and diagonal support absent from $P$. No Boolean reachability interpretation survives this basis change unless the physical entries are reconstructed first. Similarly, the spectral decomposition in TR190's symmetric setting must not be imported for an arbitrary directed, possibly defective adjacency matrix.

The private support-recovery rule remains sound: if $P$ has nonnegative integer entries and every reconstructed physical entry differs by less than $1/2$, thresholding at $1/2$ exactly recovers its support. This follows because an entry is either zero or at least one. An average coefficient error or a small spectral residual does not establish that uniform bound. For this two-label problem, exact bitsets are simpler than floating-point reconstruction and avoid the error budget altogether.

### 8.3 Algorithmic scope

The private $k$-label transform has $n2^k$ states. A per-source dynamic program costs $O(2^k(n + k|E|))$ in the stated representation and stores parent records for actual witnesses. All sources add a factor $n$ unless a separately analyzed bitset propagation is used. With $k = 2$, direct unions of first-neighbor bitsets compute the same information and permit early stopping at a successful vertex. An exact evaluator for a finite graph is not an algorithmic proof of universal success. The private reported historical batch count was read as a report claim, not adopted as a new test result.

## 9 Directed-cycle consequences and their limits

In a triangle-free oriented graph, each exact second outneighbor of $v$ is a non-neighbor: an incoming arc from it would close a directed triangle. Writing $Z(v) = V \setminus (\{v\} \cup N_1^+(v) \cup N_1^-(v))$, any successful vertex therefore satisfies

$$n - 1 = d^+(v) + d^-(v) + |Z(v)| \geq 2d^+(v) + d^-(v).$$

Conditional on universal second-neighborhood existence, triangle-freeness implies $n - 1 \geq 2\delta^+ + \delta^-$. Applying the same reasoning to the reversed graph also gives $n - 1 \geq \delta^+ + 2\delta^-$. Thus a slightly more general conditional sufficient degree test for a triangle is

$$\max\{2\delta^+ + \delta^-, \delta^+ + 2\delta^-\} \geq n.$$

It includes the source's $\delta^{\pm} \geq \lceil n/3 \rceil$ endpoint. In the $r$-regular triangle-free case it gives $n \geq 3r + 1$, so the hypothesis "directed girth exactly four" in the source can be weakened to triangle-freeness for this inequality. These are immediate conditional counting consequences, not a solution to general Caccetta–Häggkvist under a minimum-outdegree assumption alone. No conclusion for infinite or merely locally finite graphs is asserted.

## 10 Public literature refresh and formalization metadata

The public-only searches and primary-source reads were made on October 7, 2026. No private report text, path, theorem label or unpublished proposed extension was sent in a search query. The useful comparisons are narrow:

- Sullivan's AIM report, Section 6.1, gives the actual multi-label statement, its identical-two-class slice, and one *fixed* exceptional label in the degree-at-most-one special case [6].
- Seacrest's subset preprint supplies a deletion-based antecedent; the pinned manuscript explicitly proves its stronger external-image deficit rather than importing it as the older lemma [7].

- Seacrest's weighted paper gives the relevant global equivalences and the finite reversal alternative [8].
- The graph-product construction in Brantner–Brockman–Kay–Snively is an antecedent to transitive substitution, not the all-subsets estimate proved in the present report [9].
- The refreshed primary records include the 2024 ratio preprint of Huang–Peng and the 2026 minimum-outdegree-seven and dense-case preprints [10, 11, 12]. Their recorded statements provide context; their computational proofs were not replayed here. A publication record or a search result is not independent confirmation of M319.

The pinned local formalization catalogue lists

```
ComparatorChallenges/SeymourSecondNeighborhood.json
OAI.SeymourSecondNeighborhood.exists_goodVertex
OAI/Combinatorics/SecondNeighborhood/Main.lean
```

as comparator, declaration and file respectively (catalogue lines 1494–1496), and lists the manuscript at lines 142–145. The entry was inspected. No Lean source, theorem elaboration, imported axioms, compiled artifact or comparator execution was examined or replayed in this report. The local snapshot directory used for mathematical reading consists of the paper sources and catalogue material; this metadata cannot substitute for formal checking.

# 11 Advantages, limitations, and future work

The main advantage of the Ram formulation here is explicit accounting: arc-label load, Boolean propagation, distinct endpoints and external contributions are separate objects. The exact two-label slice makes applicability unusually direct. The deletion-stage margin and nonuniform block estimate add quantitative information that the source's coarse $-m + n$ bound discards. Independent clones give a clean weight transfer without confusing internal arcs with vertex masses.

The costs and limits are equally concrete. Hall allocation can fail at a successful vertex. A state transform grows exponentially with an unbounded palette. Neither invertible coordinates nor a small residual establishes endpoint expansion. The pruning lemma is the substantial source ingredient, not an automatic consequence of support computation. Exhaustive two-point relation tests are small and the larger pruning tests are seeded samples. The minimum-counterexample hypothesis is not tested by enumerating a family with no counterexamples.

Useful next work is to formalize the exact image conventions and the deletion-stage lemma separately; optimize the finite nonuniform block inequalities when their deficit data are supplied; compare the sufficient capacitated certificate with the weaker endpoint inequality; and audit the actual formal declaration and comparator at the pinned revision. An independent proof of the survivor-dependent pruning bound by purely combinatorial means would also be informative. None of these proposals licenses contraction of a private Hall-deficient palette core without an external-load and reachability preservation theorem.

# 12 Reading, computation, and delivery ledger

## 12.1 Reading coverage

The mathematical source root is

`_sources/openai-math/preprints/`
`A-proof-of-Seymours-second-neighborhood-conjecture-September-23-2026/`
`build/source/`

relative to `C:/Users/mural/OneDrive/Documents/OpenAI-722-math`. The complete files `main.tex`, all six numbered section files, `figures/pruning.tex`, and `references.bib` were read, including the full kernel-restriction and alternating-cover paragraphs. All ten manifest entries for the assigned manuscript directory, including its README, match their stored SHA-256 digests and byte counts. The standalone paper PDF is not one of those ten manifest entries; no unperformed source recompilation or PDF/source equivalence check is claimed. This verifies consistency with the supplied local manifest, not a fresh byte-for-byte download comparison against the remote Git commit.

The direct private file is

`C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/res-math/`
`TR-item-207-astra.tex`

All 650 lines were read, including definitions, local surplus, core obstructions, state transform, corrected sink reduction, multiplicity–girth hierarchy, minimal Hall witnesses, girth transfer, allocation certificate, semiring correction, complexity, limitations and bibliography. Its claims about historical tests were not rerun. The general private rainbow theorem is explicitly not claimed to be proved there, and is not upgraded here.

The three corpus maps were used only for routing and boundaries. Actual additional source passages were read as follows:

- TR190-GGRT, lines 154–329: master atom, forward and inverse definitions, core properties and scope. The discussion above uses only its conditional linear algebra and explicitly qualifies triangularization and directionality.
- `RamPatsCore50-TRs/TR-core-pat-5/TR-core-pat-5.tex`, lines 111–255: directed operators, identifiability, closure, `thm:nilpotent`, its proof, `thm:neumann`, residual and locality boundaries.
- `RamCalBooks/volumes/volume1-rev2/chapters/ ch03-local-moments-rev2.tex`, lines 191–224: polynomial closure, `thm:degree-closure`, proof and the $I - xD$ counterexample.

These three additional documents were read at the indicated passages, not in full. No further private reports mentioned in their bibliographies are silently treated as independently read or as additional evidence.

### 12.2 Exact tests and their bounded meaning

The accompanying `check_m319.py` uses integer bitsets and exact matching, writes only its assigned output directory, and hashes the input sources. It enumerates all oriented graphs on one through four labeled vertices. It tests the two-label endpoint identities, exact deletion-stage accounting, positive independent-clone formulas, and the nonuniform transitive estimate. For nonuniform substitution it uses every block-size vector from $\{1, 2, 3\}^n$ for $n \leq 3$; for $n = 4$ it uses uniform size two. Every nonempty proper subset of each tested transitive blowup is checked.

For pruning it enumerates all 16 binary relations on a two-point set, including loops, and all 16 choices of each pair family. It additionally checks 400 seeded relation/family instances at each of orders three, four and five, using seed `31920261007`. The code checks the augmented matching bound and the actual survivor charge, not just existence of a conflict-free subfamily. Two explicit failure-mode examples test walk coalescence and the nonnecessity of token matching.

**Frozen-run results.** The saved run returned `PASS`, using exact integer operations. Its counts are:

| Tested object or condition | Count |
|---|---|
| Oriented graphs, orders one through four | 760 |
| Two-label vertex identities | 3,004 |
| Independent-clone vertex identities | 70,995 |
| Transitive-subset bounds | 258,338 |
| Deletion-stage identities | 15,131 |
| Bad-to-good stages (subset of previous row) | 6,641 |
| Full-support exact cases (subset of substitution checks) | 12,368 |
| Pruning instances | 5,296 |
| Pruning with conflicts (subset of previous row) | 3,390 |
| Explicit failure-mode examples | 2 |

Overlapping counts are not added into a manufactured total. The pruning instances comprise 4,096 exhaustive two-point cases and 1,200 seeded larger cases. No claim that enumeration proves the unrestricted pruning bound or the global conjecture is made.

The native editor-open request was accepted as queued. One native compiler call failed to return during a bounded attempt and was terminated at the tool-wait level; no native compilation success is claimed. The requested on-disk PDF is exported using the existing MiKTeX helper, without installing software. The accompanying final evidence file records the actual number of formatting rounds, three TeX passes per round, diagnostics, all-page rendering and contact-sheet inspection. This report's mathematical claims do not depend on which rendering engine produced that PDF.

### 12.3 Integrity and scope of the final package

The file `checks.json` records tested-family counts and public input hash comparisons, together with observed hashes of private inputs and metadata. The separate final manifest records the frozen report, PDF, checker and evidence hashes without pretending that a self-hash can be embedded in the hashed report. All original manuscript and private corpus files remain unchanged. No catalogue, progress, queue or delivery record is modified. The report and evidence remain confidential and local.

## References


[1] OpenAI, *A proof of Seymour's second-neighborhood conjecture*, September 23, 2026. Source files at pinned commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, manuscript M319. `https://github.com/openai/math`.

[2] *Rainbow Reachability in Multi-Labeled Digraphs: Local Surplus Certificates, the Multi-Label Core, and a Ram State Transform*, TR-item-207-astra, September 7, 2026, confidential local corpus; read in full.

[3] *TR190-GGRT*, confidential local graph-transform report, Definitions `def:atom`, `def:forward`, `def:inverse`, Proposition `prop:props`, and Section `sec:scope`; selected actual passages read, as specified in the ledger.

[4] *TR-core-pat-5*, confidential local corpus, Theorems `thm:nilpotent` and `thm:neumann`, surrounding closure and directed-operator discussion; selected passages read.

[5] *Ram Calculus Books*, Volume 1 revision 2, Chapter 3, "Polynomial exactness is not polynomial closure," `thm:degree-closure`; selected actual passage read.

[6] B. D. Sullivan, *A Summary of Results and Problems Related to the Caccetta–Häggkvist Conjecture*, AIM, April 14, 2006, Sections 3 and 6.1. `https://aimath.org/WWN/caccetta/caccetta.pdf`.

[7] T. Seacrest, *Seymour's Second Neighborhood Conjecture for Subsets of Vertices*, arXiv:1808.06293v3, 2019, Lemma 4. `https://arxiv.org/abs/1808.06293v3`.

[8] T. Seacrest, *The Arc-Weighted Version of the Second Neighborhood Conjecture*, Journal of Graph Theory 78 (2015), 219–228. Preprint numbering: Theorem 1 and Proposition 2, `https://arxiv.org/abs/1212.1883v1`.

[9] J. Brantner, G. Brockman, B. Kay and E. Snively, *Contributions to Seymour's second neighborhood conjecture*, Involve 2 (2009), 387–395, Theorem 4.3. `https://msp.org/involve/2009/2-4/involve-v2-n4-p02-s.pdf`.

[10] H. Huang and F. Peng, *An improved bound on Seymour's second neighborhood conjecture*, arXiv:2412.20234v1, 2024. `https://arxiv.org/abs/2412.20234v1`.

[11] A. Sadhukhan, R. B. Sandeep and S. Sen, *A proof of Seymour's second neighborhood conjecture for oriented graphs with minimum out-degree equal to 7*, arXiv:2606.30588v1, June 29, 2026. `https://arxiv.org/abs/2606.30588v1`.

[12] J. Brukhman, *A dense-case theorem for Seymour's second neighborhood conjecture*, arXiv:2608.11530v1, August 12, 2026. `https://arxiv.org/abs/2608.11530v1`.